

Canaries in the Coal Mine? Six Facts about the Recent Employment Effects of Artificial Intelligence

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August 26, 2025

Abstract

This paper examines changes in the labor market for occupations exposed to generative artificial intelligence using high-frequency administrative data from the largest payroll software provider in the United States. We present six facts that characterize these shifts. We find that since the widespread adoption of generative AI, early-career workers (ages 22-25) in the most AI-exposed occupations have experienced a 13 percent relative decline in employment even after controlling for firm-level shocks. In contrast, employment for workers in less exposed fields and more experienced workers in the same occupations has remained stable or continued to grow. We also find that adjustments occur primarily through employment rather than compensation. Furthermore, employment declines are concentrated in occupations where AI is more likely to *automate*, rather than *augment*, human labor. Our results are robust to alternative explanations, such as excluding technology-related firms and excluding occupations amenable to remote work. These six facts provide early, large-scale evidence consistent with the hypothesis that the AI revolution is beginning to have a significant and disproportionate impact on entry-level workers in the American labor market.

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§Thanks to Nick Bloom, Joshua Gans, David Autor, Daniel Rock, Fei-Fei Li, Frank Li, Christina Langer, Sarah Bana, Cody Cook, Chris Forman, Andrew Wang, Brad Ross, Omeed Maghzian, Basil Halperin, Jiaxin Pei, Phil Trammell, Eric Bergman and participants at the Stanford Digital Economy Lab workshop for helpful feedback. We are grateful to ADP for access to the data and the Stanford Digital Economy Lab for financial support. All errors are our own.

¶Latest version: <https://digitaleconomy.stanford.edu/publications/canaries-in-the-coal-mine/>

1 Introduction

The proliferation of generative artificial intelligence (AI) has sparked a global debate about its potential impact on the labor market. This discourse, across academia, public policy, business, and popular media, spans utopian predictions of enhanced productivity, dystopian fears of widespread job displacement, and skeptical views that AI will have minimal effects on employment or productivity. Historically, technologies have affected different tasks, occupations, and industries in different ways, replacing work in some, augmenting others, and transforming still others. These heterogeneous effects suggest that there may be “canaries in the coal mine” which are harbingers of more widespread effects of AI.

There have been rapid improvements in AI capabilities in several areas. For instance, according to the most recent AI Index Report, AI systems could solve just 4.4% of coding problems on SWE-Bench, a widely used benchmark for software engineering, in 2023, but performance increased to 71.7% in 2024 (Maslej et al., 2025).¹ AI has improved on other benchmarks as well including language understanding, subject knowledge, and reasoning. At the same time, AI systems are increasingly widely adopted. According to Hartley et al. (2025), LLM adoption at work among U.S. survey respondents above age 18 reached 46% by June/July 2025.²

Given the better capabilities and widespread adoption, a central concern, amplified in recent headlines, is whether AI is beginning to supplant human labor, particularly for younger, entry-level workers in highly exposed professions like software engineering and customer service.³

Despite the intensity of this debate, empirical evidence has struggled to keep pace with technological advancement, leaving many fundamental questions unanswered. This paper confronts this empirical gap by leveraging a large-scale, high-frequency administrative dataset from ADP, the largest payroll software provider in the United States. Our sample consists of monthly, individual-level payroll records through July 2025, encompassing millions of workers across tens of thousands of firms. This rich panel structure allows us to track employment dynamics with a high degree of

¹SWE-bench is designed to evaluate the performance of large language models (LLMs) on real-world software engineering tasks. It uses a collection of GitHub issues to assess an LLM’s ability to generate code that resolves those issues.

²Similarly, Bick et al. (2024) found that in late 2024, nearly 40% of the U.S. population age 18-64 reported using generative AI, with 23% of employed respondents saying they had used generative AI for work at least once in the previous week, and 9% every work day.

³Improved productivity of workers in an occupation could lead to either reduced or increased employment, depending on, among other things, how elastic demand is for the output of those workers.

granularity, providing a near real-time view of labor market adjustments. By linking this data to established measures of occupational AI exposure and other variables, we can quantify the realized employment changes since the widespread adoption of generative AI.

This paper systematically presents six key facts that emerge from the data, offering an assessment of how the AI revolution is reshaping the American workforce.

Our first key finding is that we uncover substantial declines in employment for early-career workers (ages 22-25) in occupations most exposed to AI, such as software developers and customer service representatives. In contrast, employment trends for more experienced workers in the same occupations, and workers of all ages in less-exposed occupations such as nursing aides, have remained stable or continued to grow.

Our second key fact is that overall employment continues to grow robustly, but employment growth for young workers in particular has been stagnant since late 2022. In jobs less exposed to AI young workers have experienced comparable employment growth to older workers. In contrast, workers aged 22 to 25 have experienced a 6% decline in employment from late 2022 to July 2025 in the most AI-exposed occupations, compared to a 6-9% increase for older workers. These results suggest that declining employment AI-exposed jobs is driving tepid overall employment growth for 22- to 25- year-olds as employment for older workers continues to grow.

Our third key fact is that not all uses of AI are associated with declines in employment. In particular, entry-level employment has declined in applications of AI that *automate* work, but not those that most *augment* it. We distinguish between automation and augmentation empirically using estimates of the extent to which observed queries to Claude, the LLM, substitute or complement for the tasks in that occupation. While we find employment declines for young workers in occupations where AI primarily automates work, we find employment growth in occupations in which AI use is most augmentative. These findings are consistent with automative uses of AI substituting for labor while augmentative uses do not.

Fourth, we find that employment declines for young, AI-exposed workers remain after conditioning on firm-time effects. One class of explanations for our patterns is that they may be driven by industry- or firm-level shocks such as interest rate changes that correlate with sorting patterns by age and measured AI exposure. We test for a class of such confounders by controlling for firm-time effects in an event study regression, absorbing aggregate firm shocks that impact all workers at a

firm regardless of AI exposure. For workers aged 22-25, we find a 12 log-point decline in relative employment for the most AI-exposed quintiles compared to the least exposed quintile, a large and statistically significant effect. Estimates for other age groups are much smaller in magnitude and not statistically significant. These findings imply that the employment trends we observe are not driven by differential shocks to firms that employ a disproportionate share of AI-exposed young workers.

Fifth, the labor market adjustments are visible in employment more than compensation. In contrast to our findings for employment, we find little difference in annual salary trends by age or exposure quintile, suggesting possible wage stickiness. If so, AI may have larger effects on employment than on wages, at least initially.

Sixth, the above facts are largely consistent across various alternative sample constructions. We find that our results are not driven solely by computer occupations or by occupations susceptible to remote work and outsourcing. We also find that the AI exposure taxonomy did not meaningfully predict employment outcomes for young workers further back in time, before the widespread use of LLMs, including during the unemployment spike driven by the COVID-19 pandemic. The patterns we observe in the data appear most acutely starting in late 2022, around the time of rapid proliferation of generative AI tools.⁴ They also hold for both occupations with a high share of college graduates and ones with a low college share, suggesting deteriorating education outcomes during COVID-19 do not drive our results. For non-college workers, we find evidence that experience may serve as less of a buffer to labor market disruption, as low college share occupations exhibit divergent employment outcomes by AI exposure up to age 40.

While we caution that the facts we document may in part be influenced by factors other than generative AI, our results are consistent with the hypothesis that generative AI has begun to affect entry-level employment. We intend to continue to track the data on an ongoing basis to assess whether these trends change in the future.

Why might AI adversely affect exposed entry-level workers more than other age groups? One possibility is that, by nature of the model training process, AI replaces codified knowledge, the “book-learning” that forms the core of formal education. AI may be less capable of replacing

⁴OpenAI introduced ChatGPT in November 2022.

service representatives, and home health aides by age using data from the CPS. Though there are millions of workers employed in these professions across the US, the estimates are highly volatile, with common fluctuations of 20% or greater in estimated employment month-to-month. Figure A21 shows estimated employment changes by age and exposure quintile using the CPS, also suggesting a high degree of volatility in the estimates.

This volatility in CPS microdata reflects small sample sizes and the fact that the CPS is not stratified to target employment statistics for these demographic-occupation subgroups.²⁷ The sample size and sampling procedure of the CPS may therefore make it challenging to assess employment changes by age and AI exposure with a high degree of confidence over the time horizon considered in this paper (Chandar, 2025b; O’Brien, 2025).

Other large-scale data sources such as the American Community Survey (ACS) may offer a more reliable comparison to the ADP data, though the ACS is released with a significant lag compared to the data from ADP. We encourage comparison of our findings to results from other data sources such as the ACS upon their release.²⁸

5 Conclusion

We document six facts about the recent labor market effects of artificial intelligence.

- First, we find substantial declines in employment for early-career workers in occupations most exposed to AI, such as software development and customer support.
- Second, we show that economy-wide employment continues to grow, but employment growth for young workers has been stagnant.
- Third, entry-level employment has declined in applications of AI that *automate* work, with muted effects for those that *augment* it.
- Fourth, these employment declines remain after conditioning on firm-time effects, with a 13% relative employment decline for young workers in the most exposed occupations.

²⁷The CPS includes between 26 and 53 young software developers aged between 22 and 25 per month over our sample period. It includes between 49 and 95 young customer service representatives, and between 2 and 14 young home health aides.

²⁸The 2024 ACS 1-Year Public Use Microdata Sample is scheduled to be released on October 16, 2025.

- Fifth, these labor market adjustments are more visible in employment than in compensation.
- Sixth, we find that these patterns hold in occupations unaffected by remote work and across various alternative sample constructions.

While our main estimates may be influenced by factors other than generative AI, our results are consistent with the hypothesis that generative AI has begun to significantly affect entry-level employment.

The adoption of new technologies typically leads to heterogeneous effects across workers, resulting in an adjustment period as workers reallocate from displaced forms of work to new forms with growing labor demand (Autor et al., 2024). Such endogenous adjustment may already be happening with AI, with emerging evidence of shifts in college majors away from AI-exposed categories such as computer science (Horowitz, 2025). Past transitions such as the IT revolution ultimately led to robust growth in employment and real wages following physical and human capital adjustments, with some workers benefiting more than others (Bresnahan et al., 2002; Brynjolfsson et al., 2021).

Tracking employment trends on an ongoing basis will help determine if the adjustment to AI follows a similar pattern. Consequently, we will continue to monitor these outcomes to assess whether the trends documented in the paper accelerate in the future. Future work would benefit from better firm-level AI adoption data, which would provide sharper variation for estimating plausible causal effects of AI on employment.

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